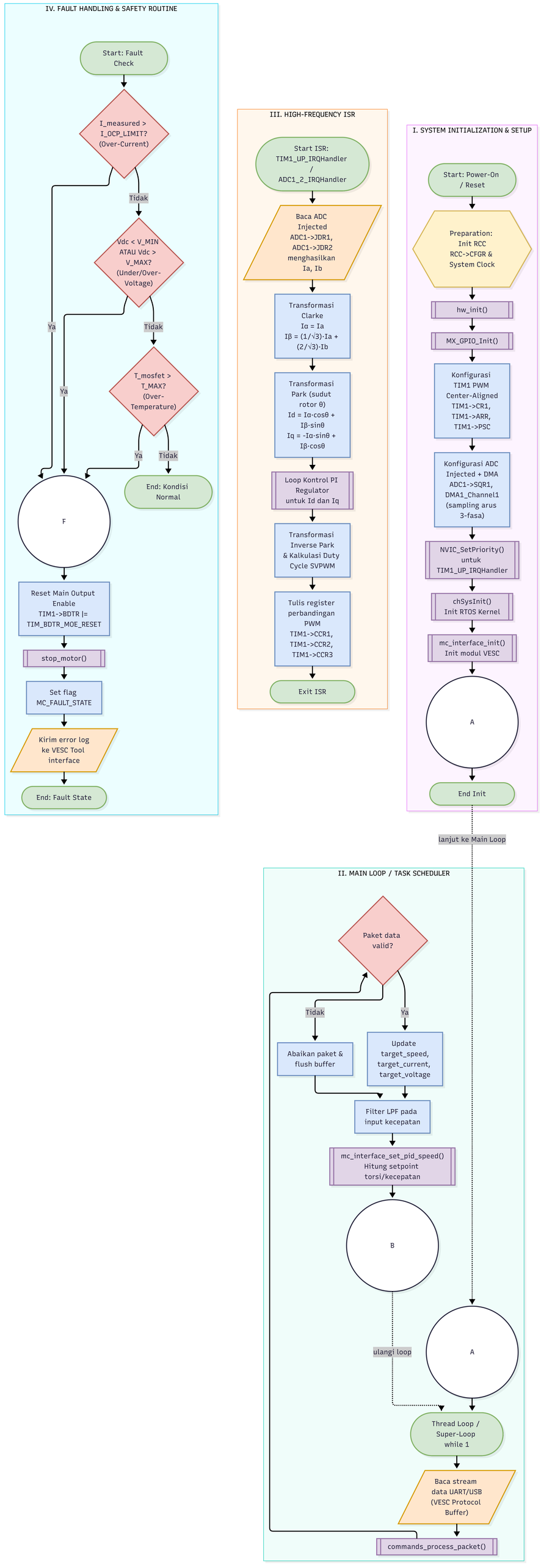
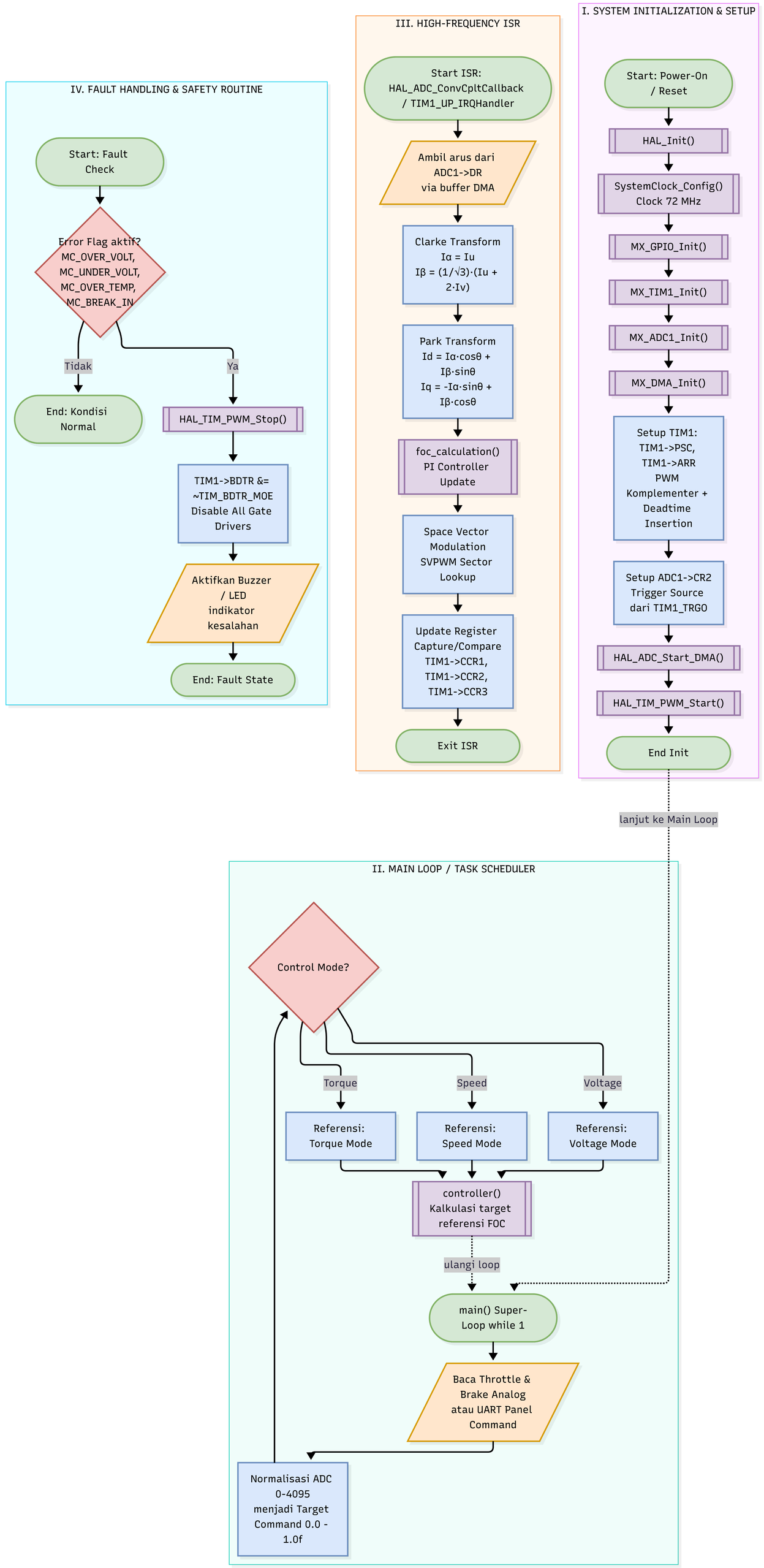
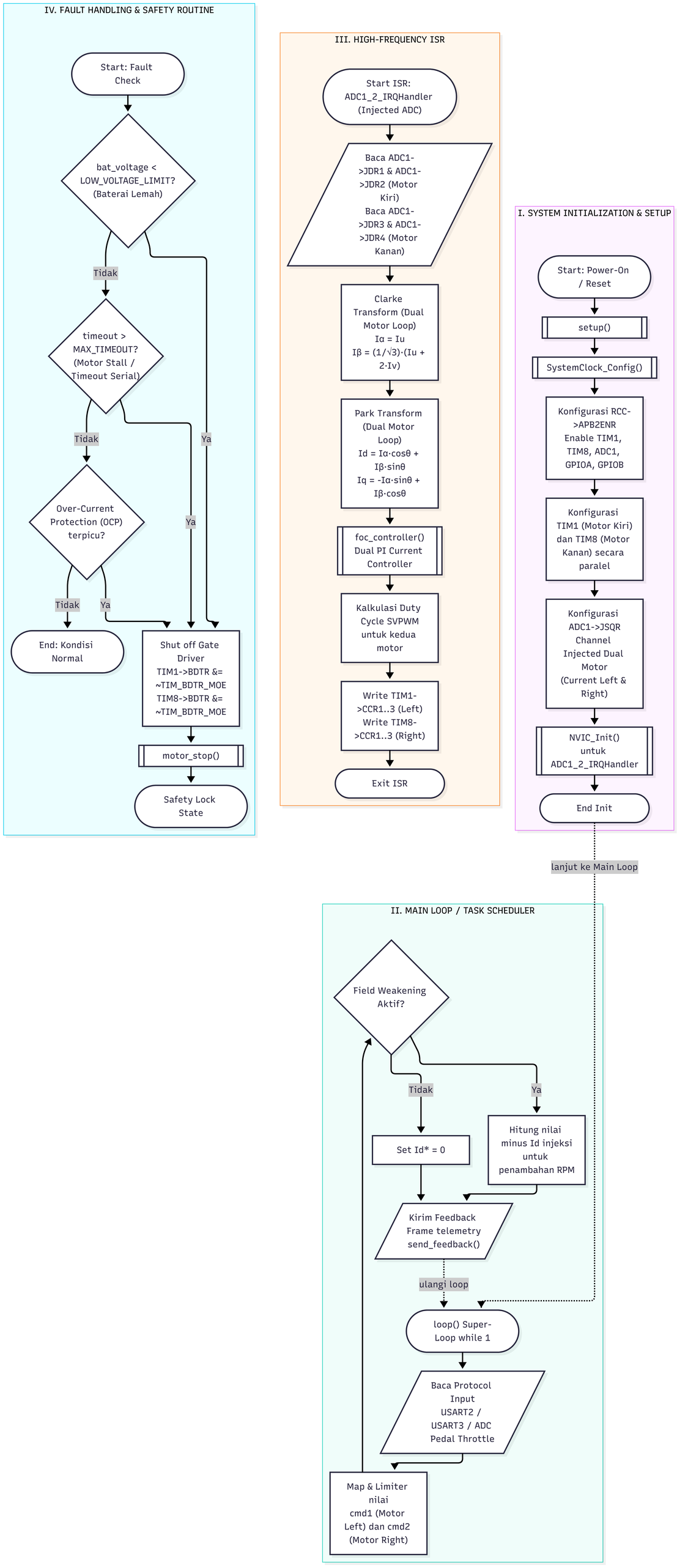

IV. FAULT HANDLING & SAFETY ROUTINE
Start: Fault Check
I_measured > I_OCP_LIMIT? (Over-Current)
Tidak
Vdc < V_MIN ATAU Vdc > V_MAX? (Under/Over-Voltage)
Ya
Tidak
Ya
T_mosfet > T_MAX? (Over-Temperature)
Ya
Tidak
End: Kondisi Normal
F
Reset Main Output Enable TIM1->BDTR |= TIM_BDTR_MOE_RESET
stop_motor()
Set flag MC_FAULT_STATE
Kirim error log ke VESC Tool interface
End: Fault State
III. HIGH-FREQUENCY ISR
Start ISR: TIM1_UP_IRQHandler / ADC1_2_IRQHandler
Baca ADC Injected ADC1->JDR1, ADC1->JDR2 menghasilkan Ia, Ib
Transformasi Clarke Iα = Ia Iβ = (1/√3)·Ia + (2/√3)·Ib
Transformasi Park (sudut rotor θ) Id = Iα·cosθ + Iβ·sinθ Iq = -Iα·sinθ + Iβ·cosθ
Loop Kontrol PI Regulator untuk Id dan Iq
Transformasi Inverse Park & Kalkulasi Duty Cycle SVPWM
Tulis register perbandingan PWM TIM1->CCR1, TIM1->CCR2, TIM1->CCR3
Exit ISR
I. SYSTEM INITIALIZATION & SETUP
Start: Power-On / Reset
Preparation: Init RCC RCC->CFGR & System Clock
hw_init()
MX_GPIO_Init()
Konfigurasi TIM1 PWM Center-Aligned TIM1->CR1, TIM1->ARR, TIM1->PSC
Konfigurasi ADC Injected + DMA ADC1->SQR1, DMA1_Channel1 (sampling arus 3-fasa)
NVIC_SetPriority() untuk TIM1_UP_IRQHandler
chSysInit() Init RTOS Kernel
mc_interface_init() Init modul VESC
A
End Init
lanjut ke Main Loop
II. MAIN LOOP / TASK SCHEDULER
Paket data valid?
Tidak
Ya
Abaikan paket & flush buffer
Update target_speed, target_current, target_voltage
Filter LPF pada input kecepatan
mc_interface_set_pid_speed() Hitung setpoint torsi/kecepatan
B
ulangi loop
A
Thread Loop / Super-Loop while 1
Baca stream data UART/USB (VESC Protocol Buffer)
commands_process_packet()


I. SYSTEM INITIALIZATION & SETUP
Start: Power-On / Reset
HAL_Init()
SystemClock_Config() Clock 72 MHz
MX_GPIO_Init()
MX_TIM1_Init()
MX_ADC1_Init()
MX_DMA_Init()
Setup TIM1: TIM1->PSC, TIM1->ARR PWM Komplementer + Deadtime Insertion
Setup ADC1->CR2 Trigger Source dari TIM1_TRGO
HAL_ADC_Start_DMA()
HAL_TIM_PWM_Start()
End Init
lanjut ke Main Loop
III. HIGH-FREQUENCY ISR
Start ISR: HAL_ADC_ConvCpltCallback / TIM1_UP_IRQHandler
Ambil arus dari ADC1->DR via buffer DMA
Clarke Transform Iα = Iu Iβ = (1/√3)·(Iu + 2·Iv)
Park Transform Id = Iα·cosθ + Iβ·sinθ Iq = -Iα·sinθ + Iβ·cosθ
foc_calculation() PI Controller Update
Space Vector Modulation SVPWM Sector Lookup
Update Register Capture/Compare TIM1->CCR1, TIM1->CCR2, TIM1->CCR3
Exit ISR
IV. FAULT HANDLING & SAFETY ROUTINE
Start: Fault Check
Error Flag aktif? MC_OVER_VOLT, MC_UNDER_VOLT, MC_OVER_TEMP, MC_BREAK_IN
Tidak
Ya
End: Kondisi Normal
HAL_TIM_PWM_Stop()
TIM1->BDTR &= ~TIM_BDTR_MOE Disable All Gate Drivers
Aktifkan Buzzer / LED indikator kesalahan
End: Fault State
II. MAIN LOOP / TASK SCHEDULER
Control Mode?
Torque
Speed
Voltage
Referensi: Torque Mode
Referensi: Speed Mode
Referensi: Voltage Mode
controller() Kalkulasi target referensi FOC
ulangi loop
main() Super-Loop while 1
Baca Throttle & Brake Analog atau UART Panel Command
Normalisasi ADC 0-4095 menjadi Target Command 0.0 - 1.0f


IV. FAULT HANDLING & SAFETY ROUTINE
Start: Fault Check
bat_voltage < LOW_VOLTAGE_LIMIT? (Baterai Lemah)
Tidak
Ya
timeout > MAX_TIMEOUT? (Motor Stall / Timeout Serial)
Tidak
Ya
Over-Current Protection (OCP) terpicu?
Tidak
Ya
End: Kondisi Normal
Shut off Gate Driver TIM1->BDTR &= ~TIM_BDTR_MOE TIM8->BDTR &= ~TIM_BDTR_MOE
motor_stop()
Safety Lock State
III. HIGH-FREQUENCY ISR
Start ISR: ADC1_2_IRQHandler (Injected ADC)
Baca ADC1->JDR1 & ADC1->JDR2 (Motor Kiri) Baca ADC1->JDR3 & ADC1->JDR4 (Motor Kanan)
Clarke Transform (Dual Motor Loop) Iα = Iu Iβ = (1/√3)·(Iu + 2·Iv)
Park Transform (Dual Motor Loop) Id = Iα·cosθ + Iβ·sinθ Iq = -Iα·sinθ + Iβ·cosθ
foc_controller() Dual PI Current Controller
Kalkulasi Duty Cycle SVPWM untuk kedua motor
Write TIM1->CCR1..3 (Left) Write TIM8->CCR1..3 (Right)
Exit ISR
I. SYSTEM INITIALIZATION & SETUP
Start: Power-On / Reset
setup()
SystemClock_Config()
Konfigurasi RCC->APB2ENR Enable TIM1, TIM8, ADC1, GPIOA, GPIOB
Konfigurasi TIM1 (Motor Kiri) dan TIM8 (Motor Kanan) secara paralel
Konfigurasi ADC1->JSQR Channel Injected Dual Motor (Current Left & Right)
NVIC_Init() untuk ADC1_2_IRQHandler
End Init
lanjut ke Main Loop
II. MAIN LOOP / TASK SCHEDULER
Field Weakening Aktif?
Tidak
Ya
Set Id* = 0
Hitung nilai minus Id injeksi untuk penambahan RPM
Kirim Feedback Frame telemetry send_feedback()
ulangi loop
loop() Super-Loop while 1
Baca Protocol Input USART2 / USART3 / ADC Pedal Throttle
Map & Limiter nilai cmd1 (Motor Left) dan cmd2 (Motor Right)